

任务一

文字说明

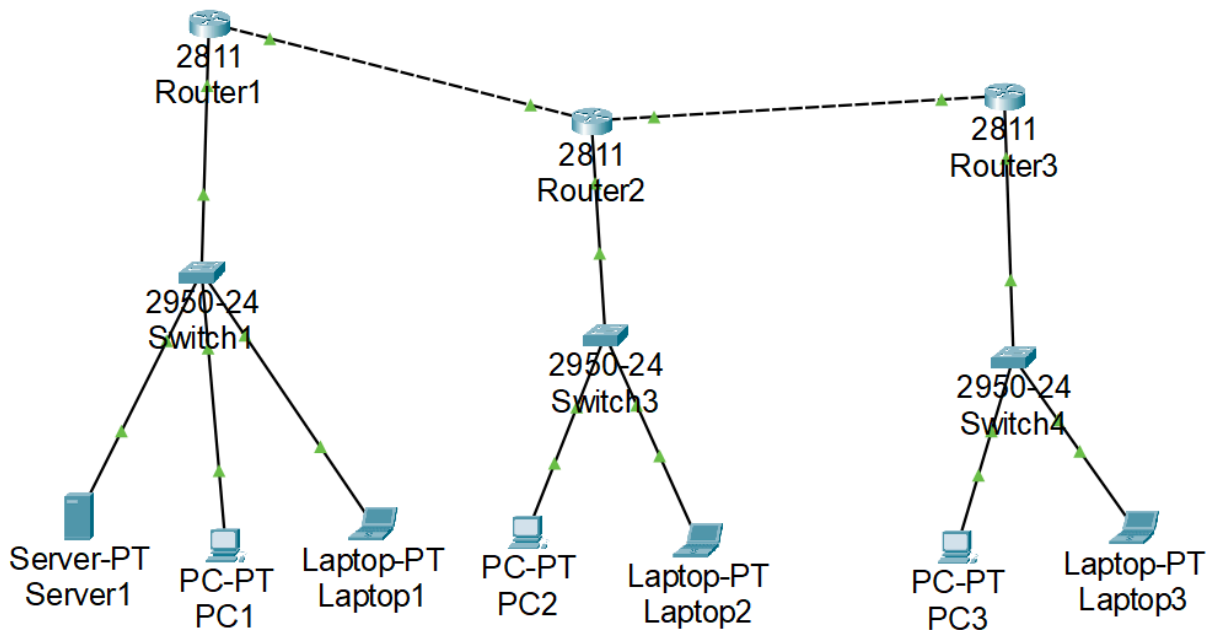
纰漏：router3的端口1，ip地址是20.0.2.1，这不属于三类私有ip地址，是公有的，会导致地址冲突，同时内部设备暴露在互联网上，很危险。

Device	Port	IP	Mask	Gateway
Router1	端口1	192.168.1.1	/24	-
	端口2	10.0.1.1	/24	-
Router2	端口1	10.0.1.2	/24	-
	端口2	10.0.2.2	/24	-
	端口3	192.168.2.1	/24	-
Router3	端口1	10.0.2.1	/24	-
	端口2	192.168.3.1	/24	-
PC1	端口1	192.168.1.2	/24	192.168.1.1
PC2	端口1	192.168.2.2	/24	192.168.2.1
PC3	端口1	192.168.3.2	/24	192.168.3.1
Server1	端口1	192.168.1.3	/24	192.168.1.1
Laptop1	端口1	192.168.1.4	/24	192.168.1.1
Laptop2	端口2	192.168.2.3	/24	192.168.2.1
Laptop3	端口3	192.168.3.3	/24	192.168.3.1

任务二

必要截图

1. 搭建：



Device Name: Router1
Custom Device Model: 2811 IOS15
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Up	--	192.168.1.1/24	<not set>	00D0.5814.9501
FastEthernet0/1	Up	--	10.0.1.1/24	<not set>	00D0.5814.9502
Vlan1	Down	1	<not set>	<not set>	0060.70A5.1B33

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1

Device Name: Router2
Custom Device Model: 2811 IOS15
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Up	--	10.0.1.2/24	<not set>	000A.41B4.6D01
FastEthernet0/1	Up	--	10.0.2.2/24	<not set>	000A.41B4.6D02
FastEthernet1/0	Up	--	192.168.2.1/24	<not set>	00E0.8FC8.879C
Vlan1	Down	1	<not set>	<not set>	0060.5CAB.438E

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router2

Device Name: Router3
Custom Device Model: 2811 IOS15
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/0	Up	--	10.0.2.1/24	<not set>	0030.A3D7.C901
FastEthernet0/1	Up	--	192.168.3.1/24	<not set>	0030.A3D7.C902
Vlan1	Down	1	<not set>	<not set>	000C.CF2D.D0CE

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router3

2. ip与rip配置 (router1为例) :

Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/0

Port Status

☒ On

Bandwidth

☒ 100 Mbps

☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex

☒ Full Duplex

☒ Auto

MAC Address00D0.5814.9501

IP Configuration

IPv4 Address192.168.1.1

Subnet Mask255.255.255.0

Tx Ring Limit10

Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/1

Port Status

☒ On

Bandwidth

☒ 100 Mbps

☐ 10 Mbps

☒ Auto

Duplex

☐ Half Duplex

☒ Full Duplex

☒ Auto

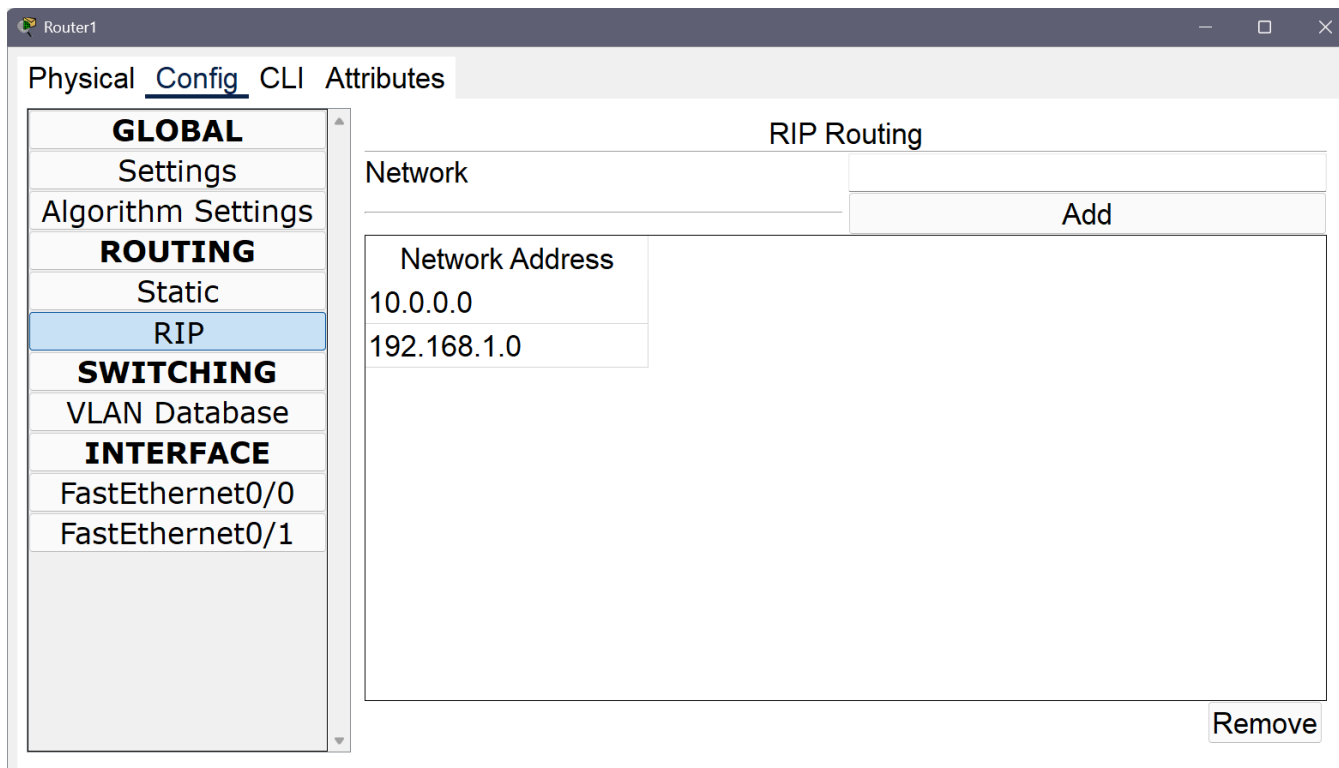
MAC Address00D0.5814.9502

IP Configuration

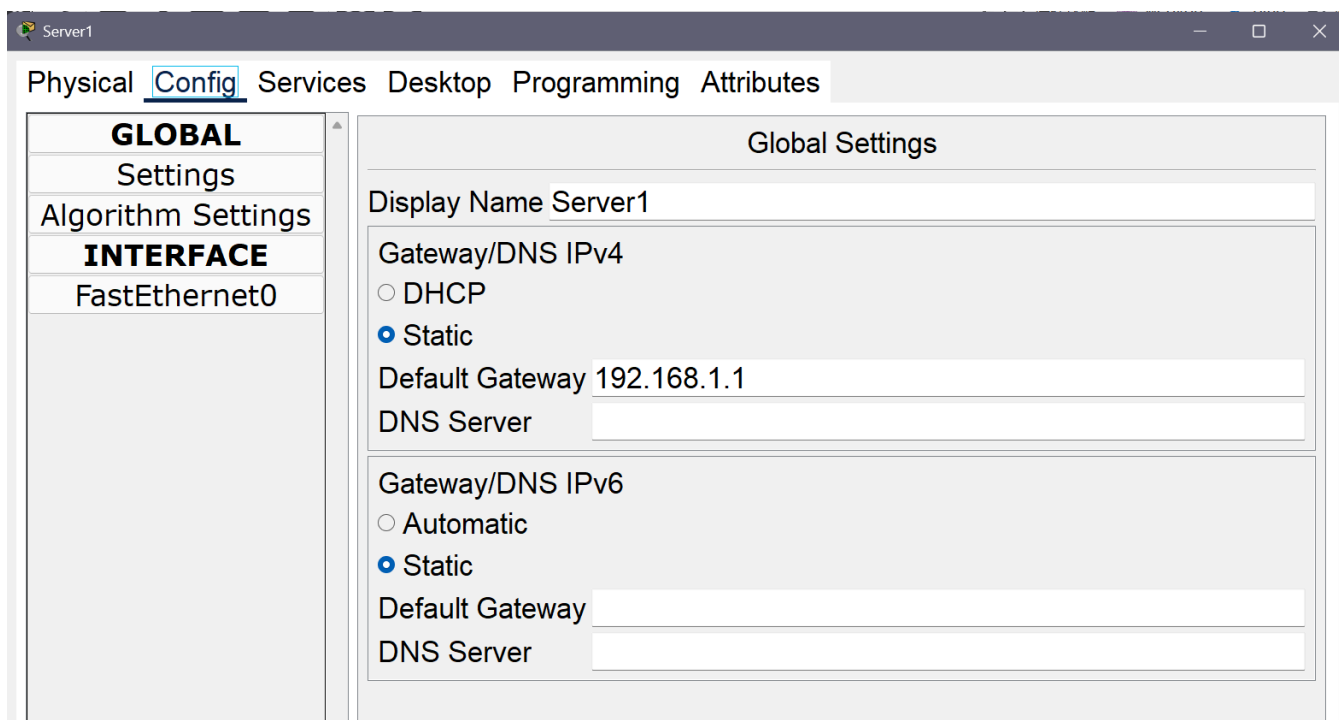
IPv4 Address10.0.1.1

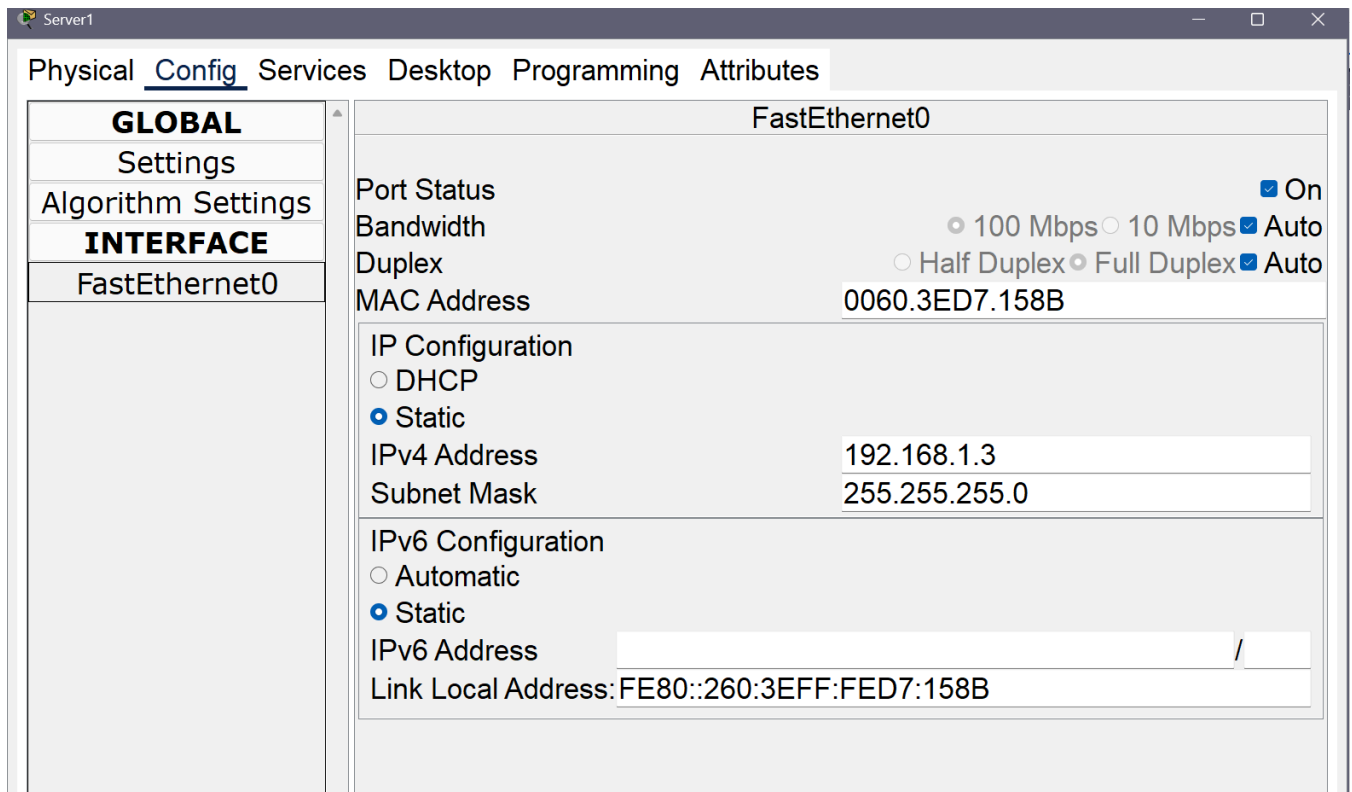
Subnet Mask255.255.255.0

Tx Ring Limit10



3. 终端的ip与gateway配置 (以server1为例) :





文字描述

先添加设备，进行连线，router2添加 NM-1FE-TX 模块，然后再用这个 Fa1/0 端口去连接交换机
然后设置router的ip和rip
然后设置终端的ip和gateway
然后将交换机的接口都 no shutdown 也就是 up

任务三

配置文件不会泄露的情况：

```
Router> enable
Router# configure terminal

# 1. 设置 Console 口令 (password1)
Router(config)# line console 0
Router(config-line)# password VENI
Router(config-line)# login
Router(config-line)# exit

# 2. 设置特权模式口令 (password2)
Router(config)# enable password VIDI

# 3. 设置 Telnet 口令 (password3)
Router(config)# line vty 0 15
Router(config-line)# password VICI
Router(config-line)# login
Router(config-line)# exit
```

```
Router(config)# end
Router# write memory <-- (保存配置)
```

配置文件可能泄露：

使用 `secret` (加密) 来代替 `password` (明文)，尤其是特权模式口令。

启用全局密码加密服务，来“混淆”那些不能用 `secret` 的口令（如console和vty）。

```
Router> enable
Router# configure terminal

Router(config)# service password-encryption
Router(config)# enable secret VENI

Router(config)# line console 0
Router(config-line)# password VIDI
Router(config-line)# login
Router(config-line)# exit

Router(config)# line vty 0 15
Router(config-line)# password VICI
Router(config-line)# login
Router(config-line)# exit

Router(config)# end
Router# write memory
```

(1) 10^6

(2) $36^6 - 10^6 - 26^6$

(3) $62^6 - 52^6 - 36^6 - 36^6 + 10^6 + 26^6 + 26^6$

(4) $62^7 - 52^7 - 36^7 - 36^7 + 10^7 + 26^7 + 26^7$

任务四

```
Router1> enable
Router1# configure terminal
Router1(config)# ip route 192.168.0.0 255.255.0.0 10.0.1.2
Router1(config)# end
Router1# write memory
```

```
Router2> enable
Router2# configure terminal
Router2(config)# ip route 192.168.1.0 255.255.255.0 10.0.1.1
Router2(config)# ip route 192.168.3.0 255.255.255.0 10.0.2.1
Router2(config)# end
Router2# write memory
```

```
Router3> enable
Router3# configure terminal
Router3(config)# ip route 192.168.0.0 255.255.0.0 10.0.2.2
Router3(config)# end
Router3# write memory
```

配置好之后，显示

Router1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.0.0/16 via 10.0.1.2

Router2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet1/0

Static Routes

Network

Mask

Next Hop

Add

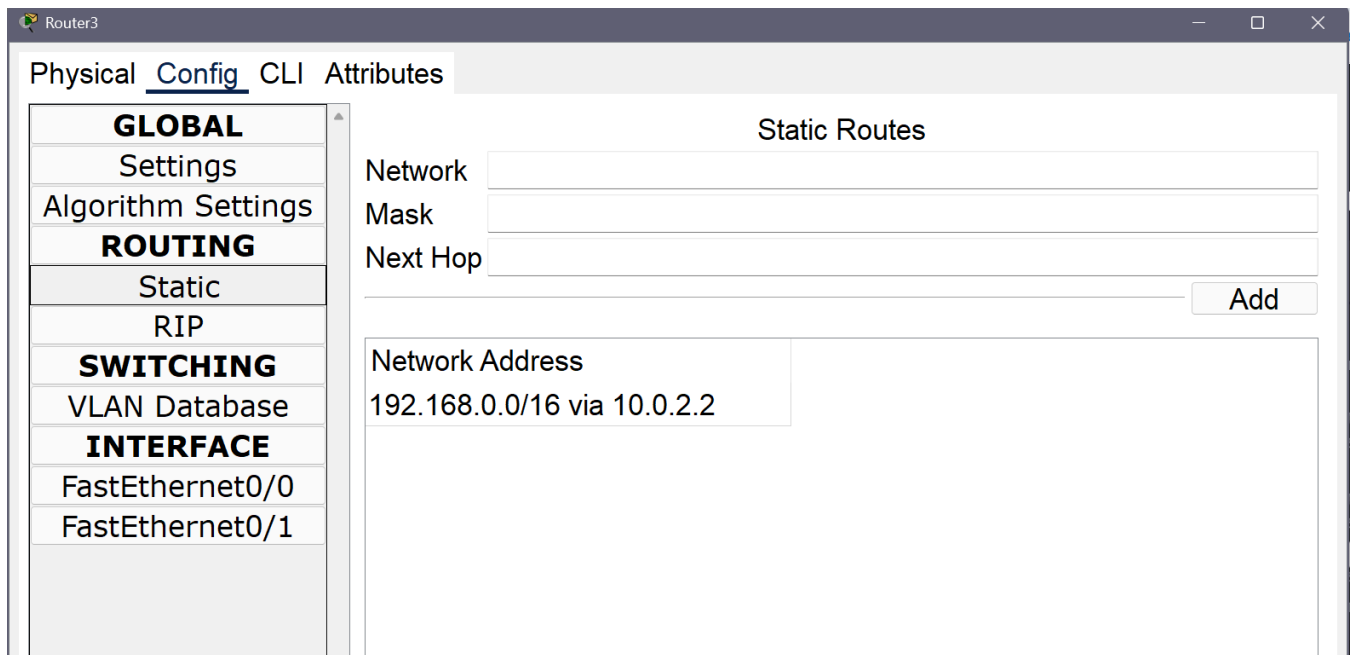
Network Address

192.168.1.0/24 via 10.0.1.1

192.168.3.0/24 via 10.0.2.1

Remove

Equivalent IOS Commands



任务五

选择 **OSPF** 路由协议维护“共和国”目前的局域网。

步骤：

为Router1和Router3添加HWIC-2T

使用 **Serial DCE** 线缆（红色Z字形，带时钟）连接 R1 的 `serial0/0/0` 和 R3 的 `Serial0/0/1`
配置接口

```
Router1(config)# interface Serial0/0/0
Router1(config-if)# ip address 10.0.3.1 255.255.255.0
Router1(config-if)# clock rate 64000
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

```
Router3(config)# interface Serial0/0/1
Router3(config-if)# ip address 10.0.3.2 255.255.255.0
Router3(config-if)# no shutdown
Router3(config-if)# exit
```

删除静态路由：

```
Router1(config)# no ip route 192.168.0.0 255.255.0.0 10.0.1.2
```

```
Router2(config)# no ip route 192.168.1.0 255.255.255.0 10.0.1.1
Router2(config)# no ip route 192.168.3.0 255.255.255.0 10.0.2.1
```

```
Router3(config)# no ip route 192.168.0.0 255.255.0.0 10.0.2.2
```

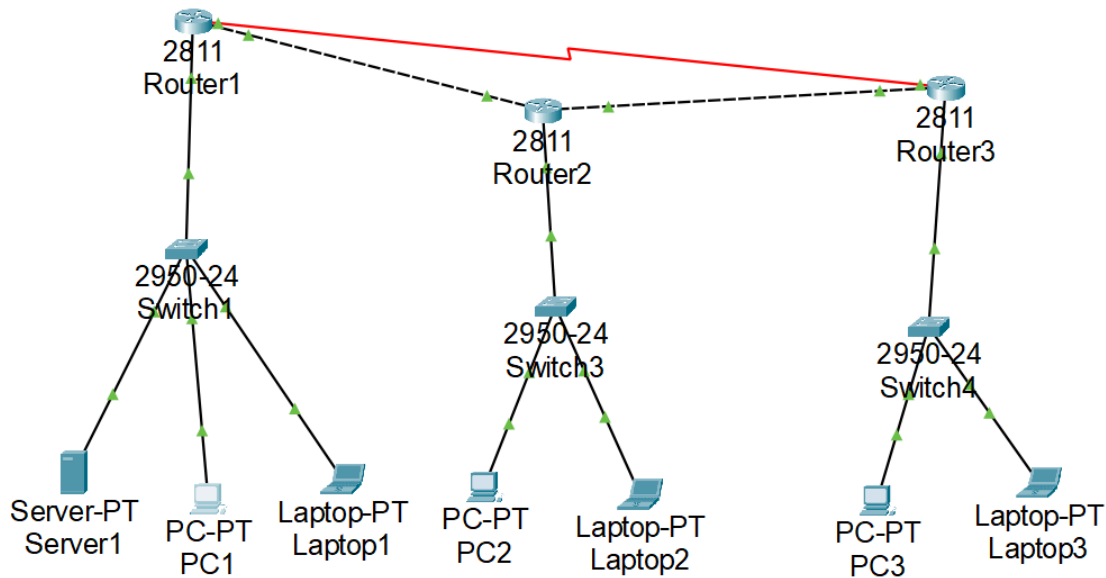
配置OSPF：


```
Router1(config)# router ospf 1
Router1(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router1(config-router)# network 10.0.1.0 0.0.0.255 area 0
Router1(config-router)# network 10.0.3.0 0.0.0.255 area 0
Router1(config-router)# end
Router1# write memory
```

```
Router2(config)# router ospf 1
Router2(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router2(config-router)# network 10.0.1.0 0.0.0.255 area 0
Router2(config-router)# network 10.0.2.0 0.0.0.255 area 0
Router2(config-router)# end
Router2# write memory
```

```
Router3(config)# router ospf 1
Router3(config-router)# network 192.168.3.0 0.0.0.255 area 0
Router3(config-router)# network 10.0.2.0 0.0.0.255 area 0
Router3(config-router)# network 10.0.3.0 0.0.0.255 area 0
Router3(config-router)# end
Router3# write memory
```

配置完之后，拓扑图是这样：



然而在PC1执行 `tracert 192.168.3.2`：

```
Cisco Packet Tracer PC Command Line 1.0
```

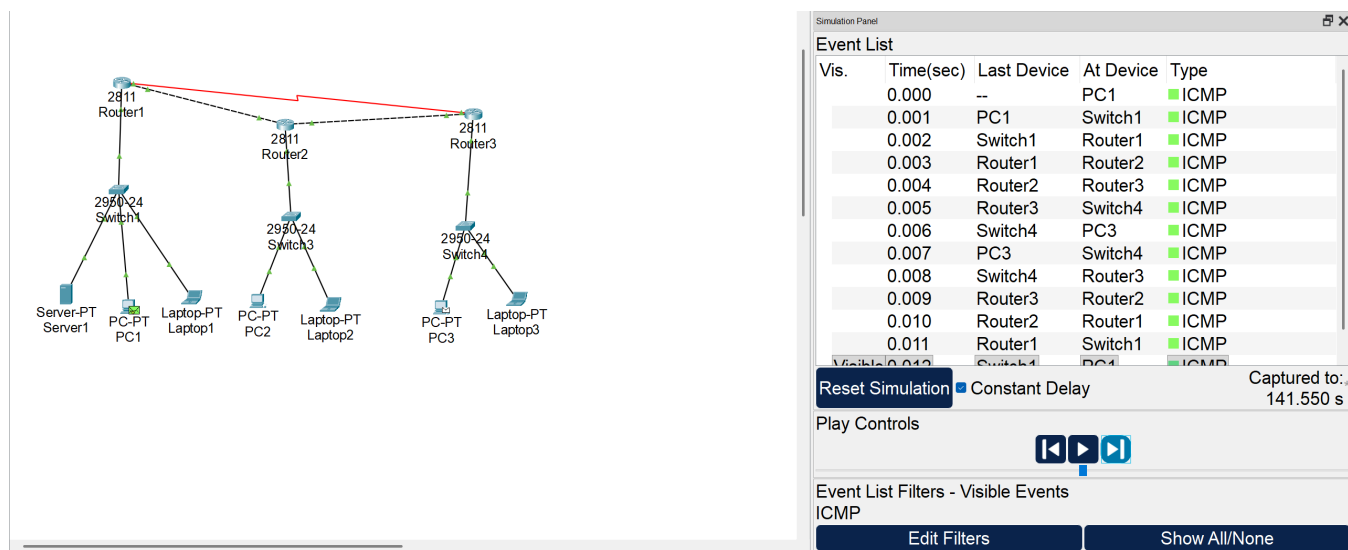
```
C:\>tracert 192.168.3.2
```

```
Tracing route to 192.168.3.2 over a maximum of 30 hops:
```

1	0 ms	0 ms	0 ms	192.168.1.1
2	0 ms	0 ms	0 ms	10.0.1.2
3	0 ms	0 ms	0 ms	10.0.2.1
4	*	0 ms	0 ms	192.168.3.2

```
Trace complete.
```

仿真化界面：



说明PC1和PC3的通信，实际路径经过了Router2，而不是新建立的R1-R3直连链路

凯撒的观点有问题，RIP协议要求子网之间的跳数不超16，而不是终端数量不超16

当前子网之间跳数最多为2，可以使用RIP作为路由协议

任务六

因为在任务三已经配置过安全密码，所以在Router1中查看运行文件：

```
Router>enable
Password:
Router#show running-config
Building configuration...

Current configuration : 1132 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$Zj.5WMmHL.bgYSBljLBWx.
enable password 7 0817656A20
!
!
!
!
!
!
ip cef
--More-- |
```

可以看到 `enable secret` 指令使用 **Type 5** 方式存储密码。这是一种基于 **MD5 (Message Digest 5)** 算法的**单向哈希**。它不会在配置文件中存储密码原文（甚至不会存储可逆的密文），而是存储一个从密码计算出的、无法反向破解的哈希值。这确保了即使配置文件泄露，攻击者也无法直接获取特权模式的密码，从而提供了高级别的安全保障。